SIMATS ENGINEERING
Department of Computer Science and Engineering

CSA15 Cloud Computing and Big Data Analytics

CO4 AT1 - Big Data Analytics Class Test Student Template

Course Code	CSA15	**Slot**	B
Course Title	Cloud Computing and Big Data Analytics	**Assessment**	CO4 AT1
Assessment Title	Big Data Analytics Class Test	**Bloom Level**	BL4 - Analyze
Faculty	Dr C. Rajesh Babu	**Employee ID**	SSETSCS415
Submission Mode	Individual digital answer template through GitHub and Google Classroom	**Total Marks**	30

Student Details

Student Name	M.GUNA SHEKAR	**Register Number**	192472345
Class / Section	C.S.E(AI)	**Slot**	B
Capstone Title	GreenHire: Cloud-Based Energy-Efficient Resume-to-Job Matcher Using NLP Skill Gap Intelligence	**Submission Date**	19-8-2026
GitHub Repository Link	https://github.com/shekar6193/CSA1512	**Google Classroom Status**	

Assessment Overview

CO4 Focus
Analyze big data characteristics, processing needs, security challenges, business drivers and domain applications. This template includes explanation notes to guide the thinking process. The final answer must be written in the student's own words.

How to use this template:

- Read each question and the explanation notes before answering.
- Use your capstone product as an example wherever relevant.
- Write concise technical answers; avoid copying the explanation notes directly.
- Use diagrams or small tables wherever they improve clarity.
- Upload the completed DOCX/PDF and supporting evidence to GitHub; share the repository link through Google Classroom.

Assessment Rubric

Criteria	Marks	Evaluation Focus	Student Check
Concept Accuracy	9	Correct technical meaning, terminology and examples.	Have I used correct big data terms?
Coverage	6	All required parts of the question are answered.	Have I answered every part?

Examples	4.5	Relevant cloud/capstone/domain examples are included.	Have I used meaningful examples?
Analysis	6	Implications, security, scalability and decision value are explained.	Have I explained why it matters?
Clarity	4.5	Answer is organized, concise and readable.	Is my answer easy to evaluate?

Code of Conduct and Student Assurance

I M.GUNA SHEKAR(19247234) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: M.GUNA SHEKAR

Question and Answer Sections

Answer all questions. Each question carries 3 marks. Use the explanation notes for direction, but write the final answer independently.

Question 1 (3 Marks)

Define Big Data and explain why traditional data processing approaches are often insufficient for big data environments.

Explanation Notes

1	Start with a precise definition: Big Data refers to datasets whose size, speed, variety or complexity exceeds the capability of conventional tools.
2	Mention why spreadsheets or single-server databases struggle: storage limits, slow queries, concurrency issues, schema rigidity and limited scalability.
3	Use one cloud/capstone-style example such as app logs, IoT feeds, transactions, images or clickstream events.
4	End with why distributed storage and parallel processing become necessary.

Student Response

1) Big Data refers to very large and complex datasets that traditional tools cannot handle efficiently.

2) It struggles with high-speed data generation and different data formats.

3) Big Data is generated from social media, mobile applications, IoT devices, and sensors.

4)1. Distributed storage stores data across multiple computers instead of one server.

2.Parallel processing divides large tasks into smaller tasks processed simultaneously.

Question 2 (3 Marks)

Explain the 5Vs of Big Data with one practical example for any two Vs.

Explanation Notes

1	List the 5Vs clearly: Volume, Velocity, Variety, Veracity and Value.
2	For at least two Vs, connect the definition to a concrete scenario instead of writing only textbook meaning.
3	Example: In a smart traffic system, velocity refers to continuous vehicle movement updates; value refers to congestion prediction.
4	Show that each V affects system design, storage, processing or analytics decisions.

Student Response

1) The 5Vs of Big Data are Volume, Velocity, Variety, Veracity, and Value.
2) Velocity refers to the continuous flow of vehicle updates in a smart traffic system, while Value refers to using this traffic data to predict congestion and improve traffic management.
3) In a smart traffic system, Velocity refers to continuous vehicle movement updates, while Value refers to using the data for traffic congestion prediction.
4) Volume affects storage capacity, Velocity affects real-time processing, Variety affects data storage and integration, Veracity affects data cleaning and quality checks, and Value affects analytics and decision-making.

Question 3 (3 Marks)

Differentiate structured, semi-structured, and unstructured data with examples.

Explanation Notes

1	Structured data has a fixed schema, such as customer table, transaction table or attendance record.
2	Semi-structured data has tags/keys but flexible structure, such as JSON, XML, API responses or log files.
3	Unstructured data has no fixed tabular form, such as images, videos, audio, PDFs, emails or social media text.
4	Use examples from one cloud-backed application to make the comparison coherent.

Student Response

1)Structured data has a fixed format and predefined schema.
2) **Semi-structured data** does not have a fixed table structure but contains **tags or keys**.
3) **Unstructured data** has no fixed tabular format or predefined schema.
4) **Structured:** Fixed schema, e.g., customer and transaction tables.
Semi-structured: Flexible structure with keys or tags, e.g., JSON and API logs.
Unstructured: No fixed structure, e.g., images, videos, PDFs, and emails.

Question 4 (3 Marks)

Explain the role of distributed file systems in big data processing.

Explanation Notes

1	State the main need: very large datasets are split and stored across multiple machines.
2	Mention core benefits: scalability, fault tolerance, parallel access and support for batch processing.
3	Use HDFS or cloud object storage as an example, but do not confuse file storage with relational database storage.
4	Connect storage to processing: data blocks can be processed in parallel by frameworks such as MapReduce or Spark.

Student Response

1) Very large datasets are split into smaller blocks and stored across multiple machines.
2) 1.It provides scalability by adding more machines when data increases.
2.It provides fault tolerance by keeping copies of data on different machines.
3) 1. HDFS stores large files by dividing them into blocks across multiple machines.
2.Cloud object storage stores data as objects in scalable cloud storage systems.
4) 1. Large datasets are divided into data blocks and stored across different machines.
2.Processing frameworks such as MapReduce and Spark process these blocks in parallel.

Question 5 (3 Marks)

List any three major security concerns in big data platforms and briefly explain each.

Explanation Notes

1	Choose three concerns such as unauthorized access, data leakage, privacy violation, weak encryption, poor audit logs or insecure APIs.
2	For each concern, explain the risk and not only the keyword.
3	Add one control where possible: role-based access, encryption, anonymization, masking, logging or monitoring.
4	Show awareness that big data combines many data sources, so security and governance become more complex.

Student Response

1) 1. Unauthorized access: Attackers or unauthorized users may access sensitive data.
   2. Data leakage: Personal or business information may be exposed accidentally or through attacks.
2) 1. Data leakage occurs when sensitive information is exposed to unauthorized people.
   2. Encryption and access control can reduce the risk.
3) 1. Encryption protects data by converting it into an unreadable form.
   2. Role-based access control allows users to access only the data they need.
4) 1. Big Data combines information from many different sources and systems.
   2. More sources increase the risk of unauthorized access and privacy violations.

Question 6 (3 Marks)

What are data appliances and integration tools? Explain their role in big data workflows.

Explanation Notes

1	A data appliance is a specialized hardware/software platform optimized for storage, processing or analytics workloads.
2	Integration tools move, transform or synchronize data between sources, storage systems and analytics layers.
3	Mention ETL/ELT, API connectors, message queues, ingestion tools or workflow schedulers as examples.
4	Explain the workflow: source data -> ingestion -> cleaning/transformation -> storage -> processing -> dashboard/decision.

Student Response

1) 1. A data appliance is a specialized hardware and software platform designed for data storage, processing, and analytics.
   2. It is optimized to handle large amounts of data efficiently.
2) 1. Data integration tools move, transform, and synchronize data between different system.
   2.They connect data sources, storage systems, and analytics platforms.
3)1. ETL/ELT tools extract, transform, and load data into storage systems.
   2.API connectors collect data from applications and external services.
4) 1. Source data is collected from applications, sensors, databases, or APIs.
   2.Ingestion brings the data into the Big Data system.

Question 7 (3 Marks)

Discuss two major challenges organizations face while implementing big data analytics.

Explanation Notes

1	Pick two strong challenges such as data quality, skill gap, infrastructure cost, privacy compliance, integration difficulty or real-time scalability.
2	Explain the cause and impact of each challenge.
3	Suggest a practical mitigation for each challenge, such as governance, cloud scaling, training, access control or phased implementation.
4	Try to connect the challenge to business impact, not only technical difficulty.

Student Response

1) **1.** A **data appliance** is a specialized hardware and software platform designed for **data storage, processing, and analytics**.
2.It is optimized to handle **large amounts of data efficiently**.
2) **1. Data integration tools** move, transform, and synchronize data between different systems.
2.They connect **data sources, storage systems, and analytics platforms**.
3)1. ETL/ELT tools extract, transform, and load data into storage systems.
2.API connectors collect data from applications and external services.
4) 1. Source data is collected from applications, sensors, databases, or APIs.
2. Ingestion brings the data into the Big Data system.

Question 8 (3 Marks)

Write short notes on any one domain application of big data analytics.

Explanation Notes

1	Select one domain: healthcare, retail, agriculture, finance, education, smart city, transport or cybersecurity.
2	Identify data sources in that domain.
3	Explain how analytics converts raw data into decisions: prediction, recommendation, alert, optimization or risk detection.
4	Mention one expected outcome such as reduced waiting time, fraud detection, crop advisory or improved user experience.

Student Response

1) 1. Healthcare analytics uses patient records, medical images, hospital data, and sensor data for analysis.
2.It helps doctors **predict diseases, provide better treatment recommendations, and improve patient care**.
2) 1. Data sources include electronic health records, medical reports, wearable devices, and laboratory results.
2. These sources provide **patient and health-related data** for analysis.
3) 1. Analytics processes healthcare data to predict diseases and identify health risks.
2.It provides **alerts and treatment recommendations** to doctors.
4) 1. Healthcare analytics can help **reduce diagnosis time and improve treatment decisions**.
2. It can provide **better patient care and improved healthcare services**.

Question 9 (3 Marks)

How do business drivers influence the adoption of big data solutions?

Explanation Notes

1	Business drivers are reasons that push an organization to adopt big data, such as cost reduction, personalization, speed, risk control or better decisions.
2	Explain how each selected driver creates a technical requirement.
3	Example: personalization requires behavior tracking and recommendation analytics; risk control requires fraud detection and monitoring.
4	Conclude that big data adoption must be justified by measurable business value.

Student Response

1) 1. Cost reduction encourages organizations to use Big Data to identify unnecessary expenses and improve resource usage.
   2. It requires data analysis and monitoring systems to find areas where costs can be reduced.

2) 1. Personalization uses customer behavior and activity data to provide individual recommendations and services.
   2. It requires behavior tracking and recommendation analytics.

3) 1. Speed helps organizations make faster decisions using real-time data processing and analytics.
   2. It requires fast data ingestion, real-time processing, and monitoring systems.

4) 1. Risk control helps organizations identify fraud, security threats, and other business risks.
   2. It requires fraud detection, data monitoring, and analytics, providing measurable business value.

Question 10 (3 Marks)

Explain the importance of big data analytics in decision making.

Explanation Notes

1	Explain that analytics helps convert raw, high-volume data into useful patterns, predictions, alerts or recommendations.
2	Differentiate descriptive, diagnostic, predictive and prescriptive decision support if space permits.
3	Use one example: dashboard for hospital beds, forecast for demand, fraud alert for bank, traffic prediction for smart city.
4	End with the idea that decisions become faster, evidence-based and scalable.

Student Response

1) 1. Descriptive analytics summarizes past data to show **what happened**.
Example: A hospital dashboard shows the **number of patients and available beds**.

2) 1. Diagnostic analytics analyzes data to find **why something happened**.
Example: A hospital can analyze patient data to find the **reason for increased waiting time**.

3) 1. Predictive analytics uses historical data to predict **future events or trends**.
Example: A bank can predict **fraudulent transactions**, while a smart city can predict **traffic congestion**.

4) 1. Prescriptive analytics recommends the **best action to take** based on data analysis.
   2. It helps organizations make **faster, evidence-based, and scalable decisions**.

Submission Checklist

Checklist Item	Status	Remarks
Student details completed.	Done / Pending	
All 10 questions answered.	Done / Pending	
Examples are connected to cloud/capstone/domain context.	Done / Pending	

Security/privacy points are included wherever relevant.	Done / Pending	
Document exported as PDF if requested.	Done / Pending	
GitHub link shared through Google Classroom.	Done / Pending	